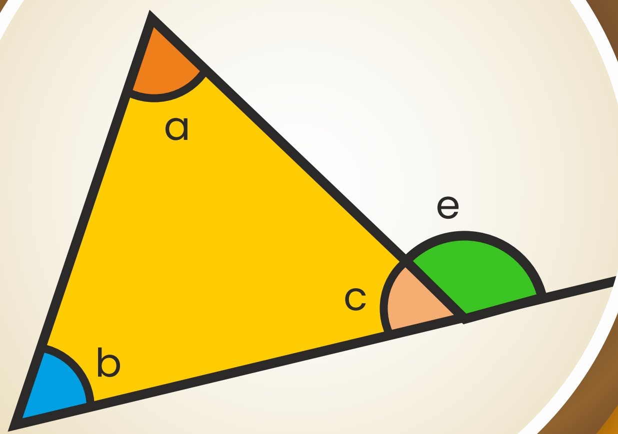
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Solution of Triangle

(English Medium)

EXERCISE (O-1)

1. In triangle ABC, if $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A.\sin B.\sin C$, then triangle is

(A) obtuse angled (B) right angled (C) obtuse right angled (D) equilateral

TS0021

2. A triangle has vertices A, B and C, and the respective opposite sides have lengths a, b and c. This triangle is inscribed in a circle of radius R. If b = c = 1 and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then R equals

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2\sqrt{2}}$

TS0022

3. In a triangle ABC, if $\angle C = 105°$, $\angle B = 45°$ and length of side AC = 2 units, then the length of the side AB is equal to

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2}+1$ (D) $\sqrt{3}+1$

TS0023

4. In a triangle ABC, if a = 13, b = 14 and c = 15, then angle A is equal to
(All symbols used have their usual meaning in a triangle.)

(A) $\sin^{-1}\frac{4}{5}$ (B) $\sin^{-1}\frac{3}{5}$ (C) $\sin^{-1}\frac{3}{4}$ (D) $\sin^{-1}\frac{2}{3}$

TS0024

5. In triangle ABC, if AC = 8, BC = 7 and D lies between A and B such that AD = 2, BD = 4, then the length CD equals

(A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$

TS0025

6. In triangle ABC, if 2b = a + c and A – C = 90°, then sin B equals
[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

TS0026

7. In a triangle ABC, $a^3 + b^3 + c^3 = c^2(a + b + c)$
(All symbol used have usual meaning in a triangle.)

Statement–1: The value of $\angle C = 60°$.

Statement –2: ΔABC must be equilateral.

(A) Statement–1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

TS0027

8. A circle is inscribed in a right triangle ABC, right angled at C. The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to

(A) 91 (B) 96 (C) 100 (D) 104

TS0028

9. In triangle ABC, If $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$ then angle C is equal to

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 30° (B) 45° (C) 60° (D) 90°

TS0029

10. In a triangle ABC, if $b=(\sqrt{3}-1)\,a$ and $\angle C=30°$, then the value of (A – B) is equal to

(All symbols used have usual meaning in a triangle.)

(A) 30° (B) 45° (C) 60° (D) 75°

TS0030

11. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c}=\sqrt{\frac{3}{2}}$, then $\angle A$ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

TS0001

12. The sides a, b, c (taken in that order) of triangle ABC are in A.P.

If $\cos\alpha=\frac{a}{b+c},\cos\beta=\frac{b}{c+a},\cos\gamma=\frac{c}{a+b}$ then $\tan^2\left(\frac{\alpha}{2}\right)+\tan^2\left(\frac{\gamma}{2}\right)$ is equal to

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

TS0032

13. AD and BE are the medians of a triangle ABC. If AD = 4, $\angle DAB=\frac{\pi}{6},\angle ABE=\frac{\pi}{3}$, then area of triangle ABC equals

(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{32}{9}\sqrt{3}$

TS0033

14. In a triangle ABC, if $(a+b+c)(a+b-c)(b+c-a)(c+a-b)=\frac{8a^2b^2c^2}{a^2+b^2+c^2}$, then the triangle is

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) isosceles (B) right angled (C) equilateral (D) obtuse angled

TS0034

15. For right angled isosceles triangle, $\frac{r}{R} =$

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\tan\frac{\pi}{12}$ (B) $\cot\frac{\pi}{12}$ (C) $\tan\frac{\pi}{8}$ (D) $\cot\frac{\pi}{8}$

TS0035

16. If K is a point on the side BC of an equilateral triangle ABC and if $\angle BAK = 15°$, then the ratio of lengths $\frac{AK}{AB}$ is

(A) $\frac{3\sqrt{2}\left(3+\sqrt{3}\right)}{2}$ (B) $\frac{\sqrt{2}\left(3+\sqrt{3}\right)}{2}$ (C) $\frac{\sqrt{2}\left(3-\sqrt{3}\right)}{2}$ (D) $\frac{3\sqrt{2}\left(3-\sqrt{3}\right)}{2}$

TS0002

17. Let ABC be a right triangle with length of side AB = 3 and hypotenuse AC = 5.

If D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is equal to

(A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$

TS0003

18. In ΔABC, if a = 2b and A = 3B, then the value of $\frac{c}{b}$ is equal to

(A) 3 (B) $\sqrt{2}$ (C) 1 (D) $\sqrt{3}$

TS0004

19. If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of expression $E = \left(\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A\right)$, is

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

TS0005

20. The ratio of the sides of a triangle ABC is $1:\sqrt{3}:2$. Then ratio of A : B : C is

(A) 3 : 5 : 2 (B) 1 : $\sqrt{3}$: 2 (C) 3 : 2 : 1 (D) 1 : 2 : 3

TS0006

21. If in a triangle sin A : sin C = sin (A – B) : sin (B – C), then a^2, b^2, c^2

(A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these

TS0007

22. In a triangle tan A : tan B : tan C = 1 : 2 : 3, then $a^2 : b^2 : c^2$ equals

(A) 5 : 8 : 9 (B) 5 : 8 : 12 (C) 3 : 5 : 8 (D) 5 : 8 : 10

TS0008

23. In an acute triangle ABC, $\angle ABC = 45°$, AB = 3 and AC = $\sqrt{6}$. The angle $\angle BAC$, is

(A) 60° (B) 65° (C) 75° (D) 15° or 75°

TS0009

24. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If a = 1, b = 3 and C = 60°, then $\sin^2 B$ is equal to

(A) $\frac{27}{28}$ (B) $\frac{3}{28}$ (C) $\frac{81}{28}$ (D) $\frac{1}{3}$

TS0010

25. In triangle ABC, If $s = 3+\sqrt{3}+\sqrt{2}$, 3B – C = 30°, A + 2B = 120°, then the length of longest side of triangle is

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 2 (B) $2\sqrt{2}$ (C) $2(\sqrt{3}+1)$ (D) $\sqrt{3}-1$

TS0011

26. In ΔABC if a = 8, b = 9, c = 10, then the value of $\frac{\tan C}{\sin B}$ is

(A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$

TS0012

27. If the sides of a triangle are $\sin\alpha, \cos\alpha, \sqrt{1+\sin\alpha\cos\alpha}, 0 < \alpha < \frac{\pi}{2}$, the largest angle is

(A) 60° (B) 90° (C) 120° (D) 150°

TS0013

28. In a triangle ABC, $\angle A = 60°$ and $b : c = \left(\sqrt{3}+1\right) : 2$ then $(\angle B - \angle C)$ has the value equal to

(A) 15° (B) 30° (C) 22.5 ° (D) 45°

TS0014

29. In triangle ABC, if $\Delta = a^2 - (b-c)^2$, then tan A =

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\frac{15}{16}$ (B) $\frac{1}{2}$ (C) $\frac{8}{17}$ (D) $\frac{8}{15}$

TS0015

30. In triangle ABC, if $\cot\frac{A}{2} = \frac{b+c}{a}$, then triangle ABC must be

[Note: All symbols used have usual meaning in ΔABC.]

(A) isosceles (B) equilateral

(C) right angled (D) isoceles right angled

TS0016

EXERCISE (O-2)

Multiple Correct Answer Type :

1. Given an acute triangle ABC such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and AB = 50. Then-

(A) centroid, orthocentre and incentre of ΔABC are collinear

(B) $\sin B = \frac{4}{5}$

(C) $\sin B = \frac{4}{7}$

(D) area of $\Delta ABC = 1200$

TS0036

2. In ΔABC, angle A is 120°, BC + CA = 20 and AB + BC = 21, then

(A) AB > AC (B) AB < AC

(C) ΔABC is isosceles (D) area of $\Delta ABC = 14\sqrt{3}$

TS0037

3. In which of the following situations, it is possible to have a triangle ABC?
(All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$ (B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$ (D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

TS0038

4. If the lengths of the medians AD,BE and CF of triangle ABC are 6, 8,10 respectively, then-

(A) AD & BE are perpendicular (B) BE and CF are perpendicular

(C) area of $\Delta ABC = 32$ (D) area of $\Delta DEF = 8$

TS0039

5. In ΔABC, angle A, B and C are in the ratio 1 : 2 : 3, then which of the following is (are) correct?
(All symbol used have usual meaning in a triangle.)

(A) Circumradius of $\Delta ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$

(C) Perimeter of $\Delta ABC = 3 + \sqrt{3}$ (D) Area of $\Delta ABC = \frac{\sqrt{3}}{8} c^2$

TS0040

6. In a triangle ABC, let BC = 1, AC = 2 and measure of angle C is 30°. Which of the following statement(s) is (are) correct?

(A) $2 \sin A = \sin B$

(B) Length of side AB equals $5 - 2\sqrt{3}$

(C) Measure of angle A is less than 30°

(D) Circumradius of triangle ABC is equal to length of side AB

TS0041

7. In a triangle ABC, if cos A cos 2B + sin A sin 2B sin C = 1, then

(A) A,B,C are in A.P. (B) B,A,C are in A.P. (C) $\frac{r}{R}=2$ (D) $\frac{r}{R}=\sqrt{2}\sin\frac{\pi}{12}$

TS0042

8. Let one angle of a triangle be 60°, the area of triangle is $10\sqrt{3}$ and perimeter is 20 cm. If a > b > c where a, b and c denote lengths of sides opposite to vertices A, B and C respectively, then which of the following is (are) correct?

(A) Inradius of triangle is $\sqrt{3}$

(B) Length of longest side of triangle is 7

(C) Circumradius of triangle is $\frac{7}{\sqrt{3}}$

(D) Radius of largest escribed circle is $\frac{1}{12}$

TS0045

9. In triangle ABC, let b = 10, c = $10\sqrt{2}$ and $R=5\sqrt{2}$ then which of the following statement(s) is (are) correct?

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) Area of triangle ABC is 50.

(B) Distance between orthocentre and circumcentre is $5\sqrt{2}$

(C) Sum of circumradius and inradius of triangle ABC is equal to 10

(D) Length of internal angle bisector of ∠ACB of triangle ABC is $\frac{5}{2\sqrt{2}}$

TS0046

10. In a triangle ABC, if a = 4, b = 8 and ∠C = 60°, then which of the following relations is (are) correct?

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) The area of triangle ABC is $8\sqrt{3}$

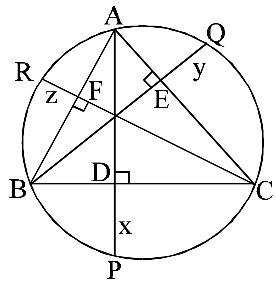
(B) The value of $\sum\sin^2 A = 2$

(C) Inradius of triangle ABC is $\frac{2\sqrt{3}}{3+\sqrt{3}}$

(D) The length of internal angle bisector of angle C is $\frac{4}{\sqrt{3}}$

TS0047

11. As shown in the figure AD is the altitude on BC and AD produced meets the circumcircle of ΔABC at P where DP = x. Similarly EQ = y and FR = z. If a, b, c respectively denotes the sides BC, CA and AB then $\frac{a}{2x}+\frac{b}{2y}+\frac{c}{2z}$ has the value equal to



(A) tanA + tanB + tanC

(B) cotA + cotB + cotC

(C) cosA + cosB + cosC

(D) cosecA + cosecB + cosecC

TS0050

Paragraph for Question 12 and 13

Let incircle of ΔABC with $\angle B = \angle C$, touches AB, BC, CA at R, P, Q respectively. If inradius 'r' is integer and perimeter of ΔABC is smallest possible integer such that $\frac{2}{AR}+\frac{5}{BP}+\frac{5}{CQ}=\frac{6}{r}$ then

12. The value of 'r' is

(A) 4 (B) 3 (C) 2 (D) 1

TS0113

13. Area of ΔABC is

(A) 15 sq. unit (B) 21 sq. unit (C) 24 sq. unit (D) 27 sq. unit

TS0114

Match the column

14. If symbols have usual notations in triangle ABC, then match the following columns :

	Column-I		Column-II
(A)	$r\,r_1r_2r_3$	(P)	rs^2
(B)	$r_1r_2r_3$	(Q)	4R cosC
(C)	$r_1+r_2-r_3+r$	(R)	$\frac{1}{2Rr}$
(D)	$\frac{1}{bc}+\frac{1}{ca}+\frac{1}{ab}$	(S)	Δ^2

TS0115

Numerical Grid Type

15. In a ΔABC with usual notations, if $3a + 3b = 4c$ then $\cot\frac{A}{2}\cot\frac{B}{2}$ is

TS0116

16. In ΔABC (with usual notations), if $r_1 = \Delta$ & $r_1 = r + r_2 + r_3$, then r is equal to

TS0117

17. In triangle ABC, a = 7, b = 6, c = 5. Incircle touches the sides BC, AC and AB at D, E and F respectively and K, L, M are the feet of perpendiculars from circumcentre to sides BC, AC and AB respectively, then the value of DK + EL + FM is

TS0106

18. If sides of a triangle ABC are given by 4,7,7 then length of the tangent drawn from vertex A to the ex-circle opposite to vertex A is equal to

TS0107

19. If f, g and h are the lengths of the perpendiculars from the circumcentre on the sides a, b and c of a triangle ABC respectively then $\frac{a}{f}+\frac{b}{g}+\frac{c}{h}=K\frac{abc}{fgh}$ where K has the value equal to ________________.

TS0108

20. In ΔABC with usual notations, if a = 3, b = 2 and $\cos(A-B)=\frac{12}{13}$, then area of ΔABC is

TS0109

EXERCISE (JM)

1. If $5, 5r, 5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to : **[JEE(Main)-Jan 2019]**

(1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $\frac{5}{4}$ (4) $\frac{7}{4}$ **TS0076**

2. With the usual notation, in ΔABC, if $\angle A + \angle B = 120°$, $a = \sqrt{3}+1$ and $b = \sqrt{3}-1$, then the ratio $\angle A : \angle B$, is : **[JEE(Main)-Jan 2019]**

(1) 7 : 1 (2) 5 : 3 (3) 9 : 7 (4) 3 : 1 **TS0077**

3. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is : **[JEE(Main)-Jan 2019]**

(1) $\frac{y}{\sqrt{3}}$ (2) $\frac{c}{\sqrt{3}}$ (3) $\frac{c}{3}$ (4) $\frac{3}{2}y$ **TS0078**

4. Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a ΔABC with usual notation. If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value :- **[JEE(Main)-Jan 2019]**

(1) (3, 4, 5) (2) (19, 7, 25) (3) (7, 19, 25) (4) (5, 12, 13)

TS0079

5. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is : **[JEE(Main)-Apr 2019]**

(1) 5 : 9 : 13 (2) 5 : 6 : 7 (3) 4 : 5 : 6 (4) 3 : 4 : 5

TS0080

6. The angles A, B and C of a triangle ABC are in A.P. and a : b = 1 : $\sqrt{3}$. If c = 4 cm, then the area (in sq. cm) of this triangle is : **[JEE(Main)-Apr 2019]**

(1) $4\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$

TS0081

7. If a ΔABC has vertices A(–1, 7), B(–7, 1) and C(5, –5), then its orthocentre has coordinates:

(1) (3, –3) (2) $\left(-\frac{3}{5}, \frac{3}{5}\right)$ (3) (–3, 3) (4) $\left(\frac{3}{5}, -\frac{3}{5}\right)$

[JEE(Main)-2020]
TS0118

8. If in a triangle ABC, AB = 5 units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is : **[JEE(Main)-2021]**

(1) $10+6\sqrt{2}$ (2) $8+2\sqrt{2}$ (3) $6+8\sqrt{3}$ (4) $4+2\sqrt{3}$

TS0119

9. In ΔABC, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of ΔABC is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of ΔABC, then the value of 2R + r (in cm) is equal to _______. **[JEE(Main)-2021]**

TS0120

10. The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for x = k, the area of the triangle is maximum, then $3k^2$ is equal to : **[JEE(Main)-2022]**

(A) 5 (B) 8 (C) 10 (D) 12 **TS0121**

EXERCISE (JA)

1. Let ABC and ABC′ be two non-congruent triangles with sides AB = 4, AC = AC′ = $2\sqrt{2}$ and angle $B = 30°$. The absolute value of the difference between the areas of these triangles is **[JEE 2009, 5]**

TS0082

2. (a) If the angle A,B and C of a triangle are in an arithmetic progression and if a,b and c denote the length of the sides opposite to A,B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C+\frac{c}{a}\sin 2A$, is -

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

TS0083

(b) Consider a triangle ABC and let a,b and c denote the length of the sides opposite to vertices A,B and C respectively. Suppose a = 6, b = 10 and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

TS0084

(c) Let ABC be a triangle such that $\angle ACB=\frac{\pi}{6}$ and let a,b and c denote the lengths of the sides opposite to A,B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are **[JEE 2010, 3+3+3]**

(A) $-\left(2+\sqrt{3}\right)$ (B) $1+\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $4\sqrt{3}$

TS0085

3. Let PQR be a triangle of area Δ with $a = 2$, $b=\frac{7}{2}$ and $c=\frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P-\sin 2P}{2\sin P+\sin 2P}$ equals **[JEE 2012, 3M, –1M]**

(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

TS0086

4. In a triangle PQR, P is the largest angle and $\cos P=\frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are)

[JEE(Advanced) 2013, 3, (–1)]

(A) 16 (B) 18 (C) 24 (D) 22

TS0087

5. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is -

[JEE(Advanced)-2014, 3(–1)]

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

TS0088

6. In a triangle XYZ, let x,y,z be the lengths of sides opposite to the angles X,Y,Z, respectively and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then-

(A) area of the triangle XYZ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$ **[JEE(Advanced)-2016, 4(–2)]**

TS0089

7. In a triangle PQR, let $\angle PQR = 30°$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) $\angle QPR = 45°$

(B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120°$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(D) The area of the circumcircle of the triangle PQR is 100π.

TS0090

8. In a non-right-angled triangle ΔPQR, let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S, the perpendicular from P meets the side QR at E, and RS and PE intersect at O. If $p = \sqrt{3}$, $q = 1$, and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(–1)]

(1) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$ (2) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2-\sqrt{3})$

(3) Length of $RS = \frac{\sqrt{7}}{2}$ (4) Length of $OE = \frac{1}{6}$

TS0091

9. Let x, y and z be positive real numbers. Suppose x, y and z are lengths of the sides of a triangle opposite to its angles X, Y and Z, respectively. If $\tan\frac{X}{2}+\tan\frac{Z}{2}=\frac{2y}{x+y+z}$, then which of the following statements is/are TRUE? **[JEE(Advanced)-2020]**

(A) $2Y = X + Z$ (B) $Y = X + Z$

(C) $\tan\frac{X}{2}=\frac{x}{y+z}$ (D) $x^2 + z^2 - y^2 = xz$ **TS0110**

10. In a triangle ABC, let $AB=\sqrt{23},\ BC=3$ and CA= 4. Then the value of $\frac{\cot A+\cot C}{\cot B}$ is_____. **[JEE(Advanced)-2021]**

TS0111

11. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R, respectively. Then which of the following statements is (are) TRUE? **[JEE(Advanced)-2021]**

TS0112

(A) $\cos P \geq 1-\frac{p^2}{2qr}$ (B) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P+\left(\frac{p-r}{p+q}\right)\cos Q$

(C) $\frac{q+r}{p}<2\frac{\sqrt{\sin Q\sin R}}{\sin P}$ (D) If $p<q$ and $p<r$, then $\cos Q>\frac{p}{r}$ and $\cos R>\frac{p}{q}$

12. Let PQRS be a quadrilateral in a plane, where QR = 1, $\angle PQR = \angle QRS = 70^\circ$, $\angle PQS = 15^\circ$ and $\angle PRS = 40^\circ$. If $\angle RPS = \theta^\circ$, PQ = α and PS = β, then the interval(s) that contain(s) the value of $4\alpha\beta\sin\theta^\circ$ is/are **[JEE(Advanced)-2022]**

(A) $\left(0,\sqrt{2}\right)$ (B) (1, 2) (C) $\left(\sqrt{2},3\right)$ (D) $\left(2\sqrt{2},3\sqrt{2}\right)$

TS0122

ANSWER KEY

Do yourself - 1

1 : **(i)** 90° **(vii)** 60° **(ix)** 45°,60°,75°

(x) a : b : c = sinA : sinB : sinC = $1 : \frac{\sqrt{3}}{2} : \frac{1}{2}$

(xii) $x \in (5, \infty)$

Do yourself - 2

2 : **(viii)** $\frac{5}{9}$ **(ix)** 25 (x) D (xi) D

Do yourself - 4

4 : **(iii)** 2 **(iv)** 30° **(v)** $\angle A = 30^\circ$, $\angle B = 135^\circ$, $\angle C = 15^\circ$

Do yourself - 5

5 : **(i)** **(a)** $\frac{3}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{\sqrt{10}}$ **(d)** $\frac{3}{\sqrt{10}}$ **(e)** $\frac{1}{3}$ **(f)** 24

(iv) $\frac{8}{15}$ **(vi)** 2 **(vii)** 3

Do yourself - 6

6 : **(i)** **(a)** 6 **(b)** $\frac{5}{2}$ **(c)** 1 **(iv)** 2 **(v)** 3 **(vi)** $\frac{9\sqrt{3}}{2}$

Do yourself - 7

7: **(ii)** 6 **(iv)** $\frac{8\pi}{5\sqrt{10+2\sqrt{5}}}$ (v) C (vi) A

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	D	A	C	C	C	A	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	D	B	C	C	B	D	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	A	C	A	C	B	D	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,D	B,C	A,C,D	B,D	A,C,D	B,D	A,C	A,B,C	A,B
Que.	11	12	13							
Ans.	A	C	D							
Q.14	A	B	C	D						
	S	P	Q	R						
Que.	15	16	17	18	19	20				
Ans.	7	1	2	9	0.25	3				

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	3	3	3	3	3	15	C

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	(a) D, (b) 3, (c) B	C	B,D	B	A,C,D	B,C,D	2,3,4	B,C	2
Que.	11	12								
Ans.	A,B	A,B								